

STA 5635 HW 4

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Problem 1

TISP variable selection method ith threshold operator:

$$\Theta(x, \lambda, \eta) = \begin{cases} 0 & \text{if } |x| \leq \lambda \\ \frac{x}{1+\eta} & |x| > \lambda \end{cases}$$

Penalized logistic regression:

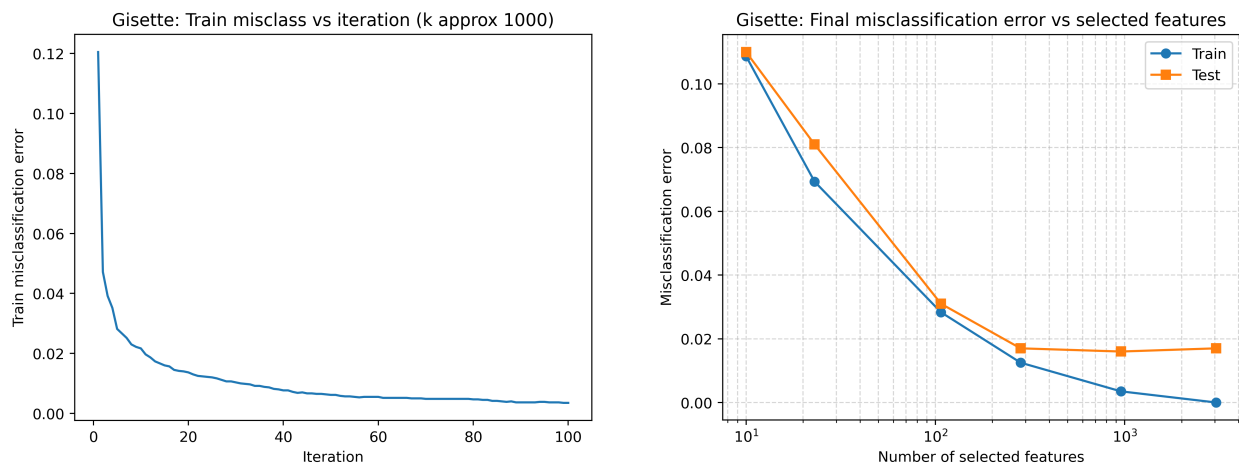
$$L(\omega) = \frac{1}{N} \sum_{j=1}^N \ln(1 + e^{-y_j^* \omega^T x_j}) + \lambda \sum_{i=1}^p p(\omega_i)$$

assuming $y_j^* \in \{-1, 1\}$ and algorithm iteration:

$$\omega^{t+1} = \Theta \left(\omega^t + \eta' X^T \left[\mathbf{y} - \frac{1}{1 + e^{-X \omega^t}} \right], \lambda \right)$$

assuming $y_j \in \{-1, 1\}$.

For $\eta = 0$ and 100 iterations, the first portion of question 1 is shown in Figure 1:



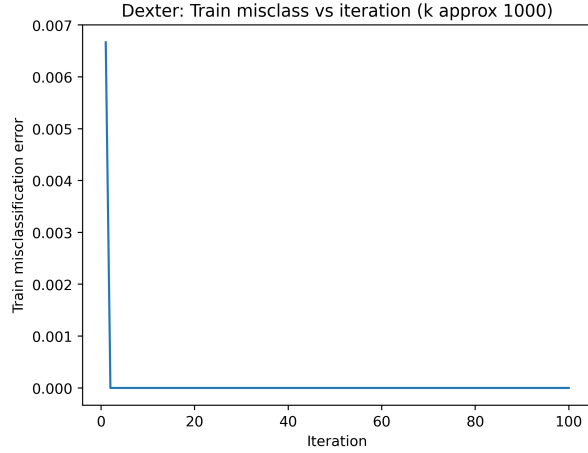
(a) Training misclassification error vs Iteration Number

(b) Train and Test Misclassification error vs Number of Selected Features

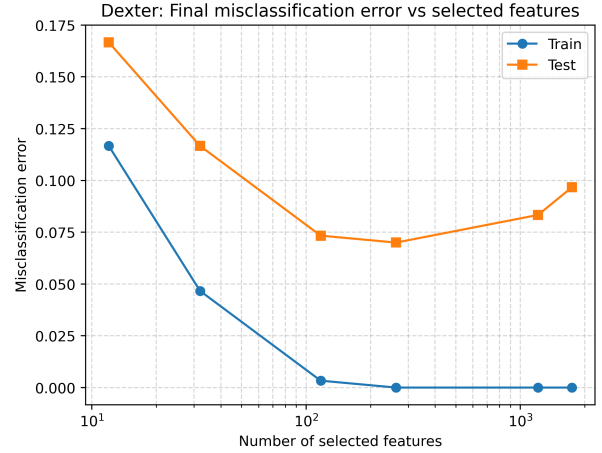
Figure 1: Train Misclassification Error vs. TISP iteration (left) and Train and Test Misclassification Error vs. Number of Selected Features (right). Gisette Dataset.

Second portion of question 1 is shown in Figure 2

The table reporting the train and test misclassification error, the test AUC, corresponding number of features selected, values of λ , and the training time are reported in



(a) Training misclassification error vs Iteration Number



(b) Train and Test Misclassification error vs Number of Selected Features

Figure 2: Train Misclassification Error vs. TISP iteration (left) and Train and Test Misclassification Error vs. Number of Selected Features (right). Dexter Dataset.

Dataset	Target Features	Selected Features	λ	Train Misclass.	Test Misclass.	Test AUC	Train Time (sec)
Gisette	10	10	0.196304	0.108667	0.110000	0.942920	1.836651
Gisette	30	23	0.138489	0.069333	0.081000	0.972464	1.785009
Gisette	100	107	0.086975	0.028333	0.031000	0.993964	1.625019
Gisette	300	282	0.054623	0.012500	0.017000	0.997420	1.622045
Gisette	1000	956	0.019179	0.003500	0.016000	0.998280	2.767586
Gisette	3000	3048	0.004229	0.000000	0.017000	0.997964	1.797582
Dexter	10	12	0.138489	0.116667	0.166667	0.932844	0.389002
Dexter	30	32	0.097701	0.046667	0.116667	0.959911	0.373019
Dexter	100	117	0.068926	0.003333	0.073333	0.975733	0.379033
Dexter	300	263	0.054623	0.000000	0.070000	0.979600	0.370544
Dexter	1000	1210	0.038535	0.000000	0.083333	0.968400	0.618003
Dexter	3000	1739	0.030539	0.000000	0.096667	0.966978	0.373005

Table 1: Train and Test Misclassification Errors, Test AUC, Number of Selected Features, values of λ , and training time, TISP algorithm.

Problem 2

FSA variable selection using logistic loss with shrinkage:

$$L_D(\beta) = \frac{1}{N} \sum_{i=1}^N \ln(1 + e^{-y_i \beta^T \mathbf{x}_j}) + s \|\beta\|_2^2$$

Inverse scheduler with parameter μ :

$$M_i = k + (p - k) \max \left(0, \frac{N^{iter} - 2i}{2i\mu + N^{iter}} \right)$$

For $s = 0.001$, $\mu = 100$, and 100 iterations, the first portion of question 2 is shown in

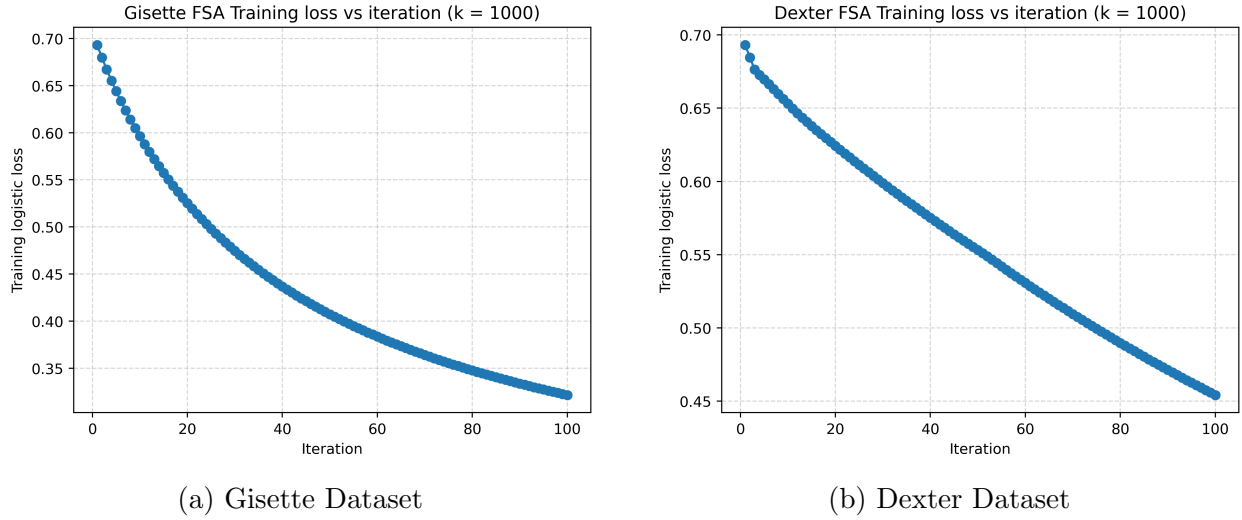


Figure 3: Plot of Training Loss vs. Iteration Number for $k = 1000$. Gisette Dataset (left), Dexter Dataset (right).

At first glance, neither datasets training loss converges, which makes sense. μ is only 100 and we only have 100 iterations; μ is usually 300 and we typically have more iterations.

The table reporting train and test misclassification errors, test AUC, and training time is shown in Table 2:

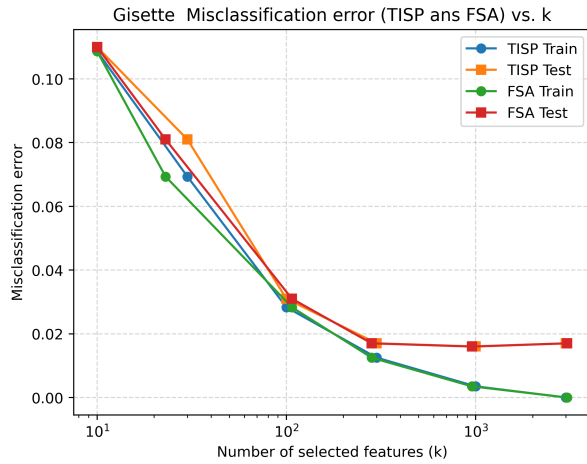
With only two exceptions (the 1000 k case), TISP appears to train faster.

We can also include the train/test misclassification error rate vs. k for the two datasets in Figure 4:

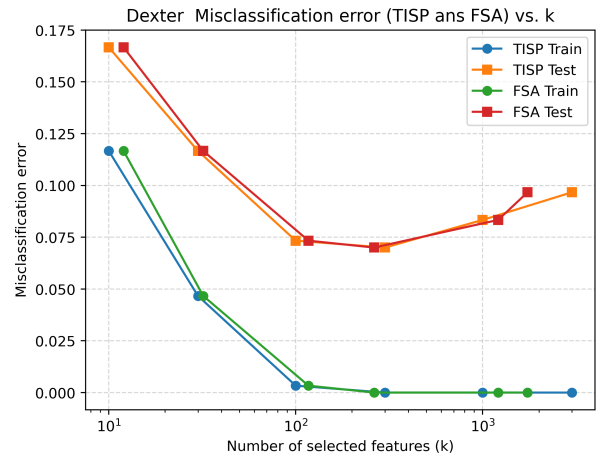
The two methods appear to be quit similar, with only slight deviations in trajectories.

Dataset	Target k	Selected	Train Miscl.	Test Miscl.	Test AUC	Time (s)
Gisette	10	10	0.139000	0.149000	0.917434	1.847994
Gisette	30	30	0.143333	0.160000	0.928632	1.857593
Gisette	100	100	0.143000	0.156000	0.935600	1.886326
Gisette	300	300	0.112167	0.112000	0.960868	1.874015
Gisette	1000	1000	0.075833	0.072000	0.979972	1.868177
Gisette	3000	3000	0.062500	0.062000	0.984572	1.876004
Dexter	10	10	0.280000	0.290000	0.917489	0.457993
Dexter	30	30	0.173333	0.193333	0.936178	0.477004
Dexter	100	100	0.056667	0.106667	0.972356	0.447000
Dexter	300	300	0.013333	0.090000	0.975378	0.451003
Dexter	1000	1000	0.003333	0.080000	0.972667	0.451003
Dexter	3000	3000	0.000000	0.123333	0.938000	0.443002

Table 2: Train and Test Misclassification Errors, Test AUC, and training time, FSA algorithm.



(a) Gisette Dataset



(b) Dexter Dataset

Figure 4: Plot of Training Loss vs. Target Features. Gisette Dataset (left), Dexter Dataset (right).